

The custom FPGA system

Engineering walkthrough, quantified parts planning and Gerald meeting questions

Prepared for Howard Wang | 18 September 2026

Nexys Video is test equipment only. The final PCB contains the FPGA chip and the selected support components, not the Nexys board.

4 ASIC carrier boards

2 ASICs each; 4,096 recording channels total

Routing / power PCB B+C

Carries individual signals and quiet ASIC supplies

Custom FPGA PCB D

XC7A200T + buffers + FX3 USB controller

KR260 receiver

USB reception, validation, buffering and 10 GbE forwarding

PC

Store the full stream; display selected/downsampled channels

We can now define a concrete architecture and many quantities. Items marked **proposed** are engineering choices to validate. Items marked **open** still need a named source, measurement or decision; they are never silently counted as zero.

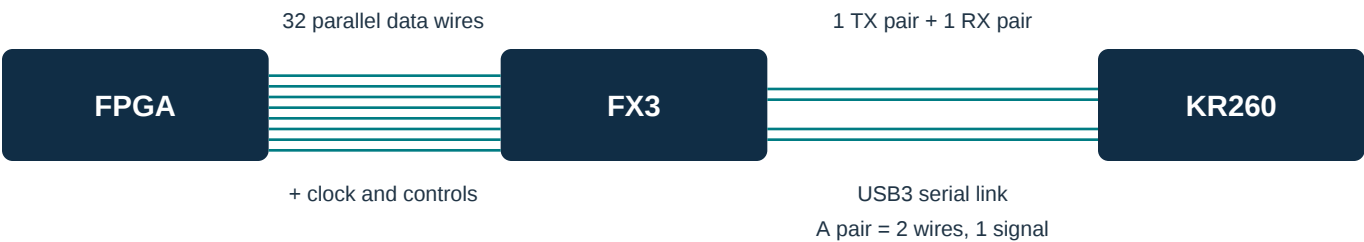
Accepted direction	Proposed baseline
8 ASICs / 4,096 channels; XC7A200T	XC7A200T-1SBG484C exact package/speed candidate
Carrier -> routing -> FPGA -> KR260 -> PC	FX3 and one USB3 link to KR260
Automatic startup; function before minimum size	256 Mbit boot flash; external recording RAM provision

Start with the data-path explanation, use the quantified hardware lists for planning, then take the numbered questions to Gerald. This is a design discussion packet, not an approved purchase order or fabrication release.

Page guide: data path 2-5; bench tools 6; quantified parts 7-12; power/source circuits 13-15; programming and work plan 16-18; Gerald questions 19-24; sources 25-27. Manufacturer and original evidence [S01-S28].

How many signal paths?

Different parts of the system use different interfaces. We do not run one cable wire for every neural channel.



Location	Physical data/timing paths	Meaning
ASICs -> routing -> FPGA	64 serial data outputs + 24 returned clock/READ/SYNC signals = 88 acquisition outputs	Eight data outputs per ASIC; each output carries many channels over time. Add command/test lines and power/ground contacts.
FPGA -> FX3, on the PCB	32 data + 1 clock + 2 address + 5 controls + 4 flags = 44 signal connections	Proposed full two-address slave-FIFO wiring. Excludes power, ground and reset/debug; fewer signals require a deliberate tie-off or firmware choice.
FX3 -> KR260, inside USB cable	1 SuperSpeed TX differential pair + 1 SuperSpeed RX differential pair	Each pair has 2 complementary wires carrying 1 serial stream. Only one pair sends our recording data toward KR260.
USB2 compatibility	1 D+/D- differential pair	Separate USB2 compatibility path, not added acquisition bandwidth.

Two wires in a differential pair do not double the data rate. TX and RX also do not add together into twice the recording bandwidth.

The five controls are SLCS, SLWR, SLRD, SLOE and PKTEND; the four flags are A-D. This count follows the AN65974 two-address example. The FPGA combines and buffers the ASIC streams. FX3 converts its parallel input into USB packets. KR260 receives a stream of packets and software recovers samples. The routing board only carries signals and supplies.

Sources: ASIC slides/reference [S01-S02]; USB3 signaling [S10]; FX3 interface [S08-S09].

Will one cable and one FX3 be enough?

It is a plausible one-link design, not a proved result. The bandwidth arithmetic supports testing it. The complete FPGA-to-PC path must pass the test before we freeze this choice.

Quantity	Calculation / limit	Result
Full-rate samples	$8 \times 512 \times 31,250$	128 million samples/s
Packed 12-bit payload	$128 \text{ million} \times 12 / 8$	192 MB/s
16-bit sample words	$128 \text{ million} \times 2$	256 MB/s
Legacy-style framing	$256 \times 132 / 128$	264 MB/s
Chosen test margin	264×1.20	316.8 MB/s
USB3 Gen1 physical signaling	$5 \text{ Gb/s} \times 8/10 \text{ coding} / 8$	500 MB/s before USB protocol overhead
FX3 32-bit GPIF at 100 MHz	$32 \times 100 \text{ million} / 8$	400 MB/s theoretical
FX3 16-bit GPIF at 100 MHz	$16 \times 100 \text{ million} / 8$	200 MB/s: insufficient

MB/s means millions of bytes per second. Gb/s means billions of bits per second. Eight bits make one byte. Do not compare these numbers without conversion.

264 MB/s uses the reference framing ratio. Our final headers, timestamps and CRC must be counted again. The older reference runs at 15,625 samples/s/channel; that does not demonstrate the 31,250-sample target.

Pass criterion: sustain the agreed stream with zero sequence gaps, correct CRC/data, bounded buffer occupancy and enough margin while KR260 forwards to PC. A cable rating by itself is not a pass.

If measured sustained throughput is below 264 MB/s, buffering cannot fix it. First optimize DMA/queues, then consider 12-bit packing or a different/multiple-link architecture. Any lower channel count or sample rate requires an explicit requirements change.

Infineon historically cites about **375 MB/s GPIF-to-USB capability**. That supports feasibility, but it is not our KR260 measurement. Internal/USB-only vendor benchmarks must not be mistaken for complete external FPGA acquisition. [S23-S24]

Sources: recording target [S01], reference framing [S02], FX3 [S08-S09], USB coding [S10]. Detailed vendor-throughput qualifications appear in the source notes.

Connector choice: count contacts, then capacity

Option	Quantity and contacts	Decision
USB3 Type-B	1 board receptacle + 1 USB3 A-to-B cable; 9 electrical contacts plus shield/mounts	Proposed simple first custom-board interface. Candidate: Wurth 692221030100.
USB-C with FX3	1 full-feature receptacle, normally 24 contacts; CC/orientation support and USB3 mux	Smaller future option. More contacts do not make FX3 faster. A USB2-only Type-C receptacle/cable is insufficient.
Micro-HDMI Type D	19 contacts; HDMI convention uses 3 data pairs and 1 clock pair	A custom link can reuse the cable only with deliberate pinout/electrical design and receiver adapter. KR260 has no native HDMI input.

For the proposed Type-B receptacle the nine electrical contacts are: **VBUS, D-, D+, GND, SSTX-, SSTX+, drain/ground, SSRX-, SSRX+**. Shell stakes are mechanical/shield connections and are not extra data lanes.

Our proposal is 1 FX3 + 1 SuperSpeed port + 1 SuperSpeed cable. This is a quantity proposal conditioned on the sustained-bandwidth test, not a promise that any USB cable will work.

Buy a cable explicitly supporting USB 5 Gb/s or better, with matching connectors. A charge-only or USB2-only cable cannot carry the target stream. Use a short bench cable first; the final cable length, flexibility and strain relief must match the experiment.

Type-C requires a correct peripheral/sink attachment implementation on CC1/CC2 and an orientation path for both SuperSpeed pairs. Its extra contacts support reversibility and other modes; FX3 still operates as a single-lane Gen1 USB device.

KR260 must receive through a SuperSpeed-capable port, negotiate at SuperSpeed and avoid competing traffic on the same controller. The proposed custom headboard is separately powered with proper VBUS detection and no backfeeding.

Concrete Type-C alternative: **1 x TE 1-2295018-2** receptacle + **1 x TI HD3SS3220** CC/orientation controller and supporting parts. This replaces the Type-B circuit. [S12, S28]

Sources: Wurth connector drawing [S11]; Type-C implementation [S12]; KR260 [S13].

Three memories with three different jobs

Memory	Proposed quantity	What it stores
FPGA boot flash	1 x 256 Mbit / 32 MiB QSPI	The FPGA program. It survives power-off. Candidate: S25FL256SAGMFI000, package/order code to verify.
Recording buffer RAM	1 external DDR device provisionally; final part and capacity open	Temporary samples while USB/host/network pauses. It loses data after power-off.
FX3 firmware storage	0 dedicated chips for host-loaded prototype; 1 compatible I2C EEPROM if autonomous boot is selected	The FX3 program. Host loading and dedicated boot memory are two explicit alternatives.

Yes: the **S25FL256S is the memory chip for FPGA startup**. It is not fast recording-buffer RAM. The XC7A200T bitstream is 77,845,216 bits (9,730,652 bytes). 128 Mbit flash fits one image; 256 Mbit provides space for normal/recovery images, with deliberate addressing and recovery configuration.

Recording buffer	Duration at 264 MB/s
Internal FPGA BRAM, all raw bits: 1,681,920 bytes	6.37 ms theoretical; usable capacity is smaller
10 ms complete output pause	Needs 2.64 MB before margin
32 MiB usable buffer	127.1 ms
64 MiB usable buffer	254.2 ms

Use the existing Nexys DDR to test buffering. The custom PCB will use a separate selected memory component if required; it will not contain the Nexys board. Reserve memory pins and power before PCB layout. Select capacity from measured stalls plus an agreed overflow policy.

One hour at the present framed rate is about **950.4 GB**. Long recording belongs on PC storage, not a small flash chip. No finite RAM can compensate for a permanently slower output link.

Sources: AMD configuration/BRAM [S04-S06], supported flash [S07], Nexys reference [S03].

Bench equipment is not the custom-board BOM

Bench item	Qty	Purpose / status
Existing Nexys Video	1 existing	Quick FPGA and DDR tests; never mounted on our custom PCB.
CYUSB3KIT-003 Explorer Kit	1 if not already available	Contains FX3, USB connector, crystal, regulators and passives. Do not buy all bare support parts just to use the kit.
FMC-to-FX3 interconnect	1 design/adaptor	Pin map, voltage, clock routing and supply isolation must be verified before connection.
SuperSpeed USB A-to-B cable	1 per active link	Correct USB3 contacts and negotiated SuperSpeed operation.
KR260	1 system receiver	USB capture plus configured network forwarding.
External FPGA JTAG programmer	1 shared bench tool	Connects to 1 header on each custom PCB; no separate USB programming receptacle required there.
Headboard power source	1	12 V / 2 A planning rating for stated envelope; current-limited bring-up.
KR260 power source	1 separate	Use its specified supply; excluded from headboard budget.
PC with fast storage and suitable network interface	1	Recording, validation and selected-channel display.
10 GbE physical link	1 link	Either 1 compatible SFP+ DAC, or 2 compatible optical modules + 1 matching fiber; verify endpoints.

Before connecting the adapter: Nexys FMC VADJ defaults to 1.2 V. Set and verify a voltage compatible with the chosen FX3 I/O configuration; do not assume factory settings match. Oscilloscope, logic analyzer, DMM/current measurement and dummy loads are shared lab tools. Confirm existing inventory before purchasing.

The next purchasing step is a checked kit/adaptor list for the bench or a schematic-derived BOM for fabrication. The two lists must not be mixed.

Sources: Nexys [S03], Explorer Kit [S22], KR260 [S13]. No availability or lab inventory is assumed beyond the supplied existing Nexys context.

Shopping-style list: custom FPGA PCB

Quantities are **per one FPGA PCB**. P = proposed count/candidate; O = open item. Existing Nexys and KR260 boards are not components in this BOM.

ID	Qty	Part / specification	Purpose and release condition
F01	1 P	XC7A200T-1SBG484C	Main FPGA; validate exact grade/package, pins, timing and availability.
F02	1 P	S25FL256SAGMFI000; 256 Mbit QSPI	FPGA boot flash; verify package, sector layout and Vivado configuration.
F03	1 P	CYUSB3014-BZXC FX3	USB3 bridge; 32-bit interface, 121-ball BGA. BZXI is an alternative temperature grade, not a second chip.
F04	1 P	ADP5052ACPZ-R7	Four buck channels: 1.0/1.8/1.5/3.3 V. Support list on next page.
F05	1 P	TPS62160DSGR	FX3 1.2 V rail; 1 A class. Verify transient response.
F06	1 P	100 MHz active oscillator	FPGA reference clock; output voltage, jitter and package still need exact selection.
F07	1 P	19.2 MHz FX3 crystal	Separate USB-controller reference; confirm crystal parameters and load components.
F08	1 P	Wurth 692221030100	9-contact USB3 Type-B receptacle; verify body height/board placement.
F09	1 P	JTAG programming header	External programmer access; choose a standard pinout and matching cable.
F10	1 P	External DDR component, exact part O	Provisional recording buffer. Capacity/controller/pins not frozen.
F11	0 or 1 O	FX3 boot I2C EEPROM	0 for host loading; 1 if autonomous FX3 startup is selected.
F12	O	Routing connector(s), exact contact count O	Count full signals, grounds and rails before selecting mating parts.

F06/F07 are real physical timing components. Their clock signals also require correct PCB routing and FPGA/FX3 configuration. Exact oscillator/crystal order codes remain open; do not order by frequency alone.

Sources: device and package [S03-S09], USB connector [S11], power converters [S14-S15].

Shopping-style list: support components

These counts describe the selected circuit blocks, not a finished total BOM. Values and quantities marked open must be resolved from the chosen reference circuit and final schematic; they are not zero.

ID / block	Qty baseline	Part and reason
P01: ADP5052 buck inductors	4	1 output inductor per buck channel; choose inductance, saturation current and DCR.
P02: ADP5052 low-side MOSFETs	2 functions / 1 package	1 x Si7232DN-T1-GE3 dual MOSFET candidate supplies both CH1/CH2 switches; verify transients and losses.
P03: TPS62160 output inductor	1	Additional FX3 1.2 V converter. Total power inductors in this topology: 5.
P04: adjustable output feedback	10 resistors P	2 per each of 5 adjustable buck outputs, assuming no alternative fixed-output implementation. Values calculated per rail.
P05: converter compensation / bootstrap / soft-start / enable / frequency	Counts on detailed ADP page	4 bootstrap caps; 4 compensation R+C pairs; 4 optional pole caps; 1 RT; 2 ILIM; 4 soft-start and 8 enable positions, plus mode/status.
P06: converter input/output capacitors	Open	Depend on stability, transient load, ripple and overlap with the FPGA PDN. Avoid accidental double counting.
P07: FPGA supply capacitors	79	Exact grouped reference list on next page, under the stated bank assumptions.
P08: FPGA oscillator bypass	1 capacitor P	Local bypass; value follows oscillator datasheet. Separate from the 79.
P09: FX3 crystal load network	2 capacitors P; series resistor footprint conditional	Values from crystal load capacitance, FX3 requirements and PCB parasitics; verify reference circuit.
P10: USB TX AC coupling	2 capacitors P	One per FX3 SuperSpeed TX conductor; value/placement per FX3 reference.
P11: ESD protection, FX3 bypass/filtering, flash bypass	Open count	Every supply domain/interface must be covered; exact arrays/package and grouping determine physical part count.
P12: reset/boot resistors, buttons, LEDs/test points	Open count	Choose explicit boot/reset scheme. Minimum functions explained on page 12.

Why some totals are open: there is not yet a frozen custom schematic. Reporting an exact complete resistor/capacitor count would be invented. Each open row has a closure task in the checklist; final purchasing waits for that schematic-derived BOM.

Converter sources [S14-S15]; FPGA network [S16]; FX3 hardware [S08]. See support-parts audit for reference-specific detail.

The 79 FPGA capacitors: exact baseline

Supply domain	Quantity / value
VCCINT	1 x 680 uF + 12 x 4.7 uF + 14 x 0.47 uF
VCCBRAM	1 x 100 uF + 3 x 0.47 uF
VCCAUX	1 x 47 uF + 3 x 4.7 uF + 5 x 0.47 uF
VCCO bank 0	1 x 47 uF
Each of 6 user-I/O banks	2 x 4.7 uF + 4 x 0.47 uF
Shared VCCO bulk	2 x 47 uF for the illustrated two voltage groups

Proposed bank grouping: banks 13/14 at 3.3 V for GPIF/configuration-compatible functions; banks 15/16/34/35 at provisional 1.5 V for ASIC/possible memory functions. Bank 0 also needs the proper configuration supply. Shared bulk rule: one 47 or 100 uF part per up to four banks at the same voltage.

Value / qty	Reference body / type	ESL maximum / ESR range
680 uF x 1	2917 / D / 7343 tantalum	2 nH / 5 < ESR < 40 milliohm
100 uF x 1	1210; tantalum or X5R/X7R	1 nH / 1 < ESR < 40 milliohm
47 uF x 4	1210 X5R/X7R ceramic	1 nH / 1 < ESR < 40 milliohm
4.7 uF x 27	0805 X5R/X7R ceramic	0.5 nH / 1 < ESR < 20 milliohm
0.47 uF x 46	0603 X5R/X7R ceramic	0.5 nH / 1 < ESR < 20 milliohm

AMD reference voltage ratings are 2.5 V for 680/100 uF and 6.3 V for the smaller values. Historical example parts, respectively: T530X687M006ATE018; GRM32ER60J107ME20L; GRM32ER70J476ME20L; GRM21BR71A475KA73; GRM188R70J474KA01. These are reference examples, not newly verified purchase approvals.

79 is the FPGA reference network only. Add regulator and peripheral requirements. Check DC-bias derating, tolerance, ESR/ESL, mounting inductance and regulator stability before substitutions.

Source: AMD UG483 Table 2-2 and Table 2-5 [S16]. Core/BRAM sharing a 1 V regulator does not automatically remove either network.

ADP5052: count every support position

Proposed component: **1 x ADP5052ACPZ-R7**. Start channel mapping with CH1=1.0 V, CH2=1.8 V, CH3=1.5 V and CH4=3.3 V. The internal 200 mA LDO is unused. These are reference-derived positions; values and population still need the actual circuit calculation.

Item	Qty	Value / condition
Power inductors	4	Values/order codes open; calculate for frequency, ripple, peak current and temperature.
Low-side switching package	1	Si7232DN-T1-GE3 dual N-MOSFET = 2 transistor functions; verify 20 V transient margin.
Bootstrap capacitors	4	0.1 uF each, with appropriate dielectric/rating.
Local input capacitors	4	10 uF nominal per buck position; 25 V candidate rating for 12 V feed, check derating.
VREG / VDD bypass	2	1 uF each.
Feedback resistors	8	2 per output; values set from 0.8 V reference and tolerance budget.
Compensation resistors	4	One per control loop; values open.
Compensation capacitors	4 + 4 optional	4 main loop capacitors; up to 4 optional high-frequency pole capacitors.
Frequency resistor	1	RT; value open after minimum-on-time/frequency calculation.
Current-limit positions	2	CH1/CH2; CH1 22 kohm proposed for 4 A mode; second selection open.
Soft-start divider positions	4 reserved	2 each on SS12/SS34, or 0 fitted if appropriate hardwired setting is chosen.
Enable/UVLO divider positions	8 reserved	2 per channel; fitted count follows selected sequencing design.
Power-good pull-up	1	10 kohm proposed; compatible logic voltage required.
MODE/SYNC position	1 reserved	0 ohm strap or suitable pull resistor, mode open.
Local output capacitor positions	6 provisional	2+2+1+1 planned positions; values and overlap with 79-cap FPGA network must be reconciled.

Soft-start is selected through resistor/tie settings for supported 2/4/8 ms options, not external timing capacitors. The earlier 5 ms charge-current example was arithmetic, not a selectable ADP5052 setting. Do not add six local output capacitors blindly to the 79: check the complete per-rail capacitance and stability.

Sources: ADP5052 Figure 55 and design sections [S14]; Vishay dual-MOSFET [S20].

FX3 regulator and clock-part counts

TPS62160 1.2 V / 1 A reference: TI Figure 39 gives a concrete seven-part starting circuit. This is separate from FX3 bypass/filtering and may gain enable circuitry in our sequenced design.

Item	Qty	Reference value / candidate
Regulator IC	1	TPS62160DSGR, adjustable WSON version
Inductor	1	2.2 uH; TDK VLF3012ST-2R2M1R4 is a reference candidate, current checks required
Input capacitor	1	10 uF, 25 V, 0805 reference
Output capacitor	1	22 uF, 6.3 V, 0805 reference
Feedback top / bottom resistors	1 + 1	75 kohm / 150 kohm
Power-good pull-up	1	100 kohm
External MOSFET / bootstrap / compensation	0 / 0 / 0	These are not external parts in this reference circuit

Clock circuits: these are physical components as well as timing signals.

Item	Qty	Selection and next check
FPGA active oscillator	1	100 MHz SiT8008B-family candidate; exact voltage/package/stability/OE order code open
Oscillator bypass	1	At least 0.1 uF for the candidate; select exact part
Oscillator series / OE resistor positions	1 + 1	Series initially 0 ohm or DNP; OE <=10 kohm if required, or choose NC option
FX3 crystal	1	19.2 MHz, +/-100 ppm or better and other FX3 crystal limits
FX3 crystal load capacitors	2	Values from crystal load, pin/PCB capacitance; do not guess 18 or 22 pF
FX3 FSLC strap positions	3	Propose FSLC=000 with 0 ohm straps; direct copper ties require 0 physical resistors

A 3.3 V FPGA oscillator needs a compatible clock-capable FPGA pin. Do not copy the Nexys clock pin if our bank is now 1.5 V. Independent FPGA and FX3 references allow FX3 startup without depending on configured application logic.

Sources: TPS62160 Figure 39 [S15]; SiTime oscillator [S21]; FX3 crystal/clock requirements [S08].

Routing PCB: quantities and evidence

Item	Quantity per routing assembly	Purpose / next source
Carrier-side interfaces	4 P	One per two-ASIC carrier if that is the actual carrier interface. Gerald supplies exact carrier revision and mating connectors.
FPGA-facing connector(s)	Open	88 acquisition outputs plus controls/test/power/grounds; we create the contact inventory.
Low-voltage LT3042 regulators	7 per shared group P	Group count g is open; total = 7g. Possible g=1/4/8 means 7/28/56, not three simultaneous purchases.
15 kohm SET resistors	5g	Sets five nominal 1.5 V rails per group.
16 kohm SET resistors	1g	Sets nominal 1.6 V clock-generator rail.
7.5 kohm SET resistors	1g	Sets nominal 0.75 V common-mode reference.
LDO input/output/SET capacitors	Open for final circuit	Count and effective values from validated LT3042 circuit, not a generic per-chip guess.
High-voltage supply components	Open; do not order yet	Gerald confirms actual stimulation supplies and validated schematic.
Current links/test points/clock buffers	Open exact count	Determine after rail groups, fanout and probe access are defined.

Gerald needs to tell us which rails are already present on each carrier and whether sharing is allowed. We then measure rail current and recording noise with 1 ASIC, 2 ASICs and the intended larger grouping. A regulator datasheet tells us what the regulator supports; it cannot tell us the proprietary ASIC current or crosstalk.

Practical sharing test: compare per-rail voltage/ripple and input-referred recording noise with neighboring ASICs idle and active. Include clock loading, startup, gain changes and permitted stimulation. Use an agreed acceptable noise/voltage limit, not just whether the board powers up.

Ask Gerald for the current tested carrier/LDO schematic and actual assembled part numbers. The old files contain a regulator-package mismatch that must be reconciled before copying footprints.

Sources: supplied LDO schematic [S17], LT3042 [S18], ASIC slides [S01]. Questions G01-G16 identify the requested evidence.

Power: ratings versus actual consumption

Rail	Proposed rating	Consumers / caveat
1.0 V	4 A	FPGA core + BRAM
1.8 V	1.2 A	FPGA auxiliary / filtered analog support where required
1.5 V	1.2 A	ASIC-facing FPGA I/O; DDR requires updated load budget
3.3 V	1.2 A	Flash/GPIF/FX3 support plus nested 0.5 A ASIC allocation
1.2 V	1 A	FX3 core/USB PHY; transient response still needs validation

Provisioned output power = $1 \times 4 + 1.8 \times 1.2 + 1.5 \times 1.2 + 3.3 \times 1.2 + 1.2 \times 1 = \mathbf{13.12\ W}$. With assumed 85% efficiency and 25% input margin: $13.12 / 0.85 / 1.25 = \mathbf{1.608\ A}$. A **12 V, 2 A planning source** covers that stated envelope. KR260, stimulation supplies and later added loads are excluded.

13.12 W is the sum of selected rail capacities, not a claim that the headboard will consume or dissipate 13.12 W.

The ASIC slide gives 17.24 uW/channel nominal recording power, excluding LDOs: $4,096 \times 17.24\ \mu\text{W} = \mathbf{70.615\ mW}$. Equivalent current at 1.5 V would be 47.08 mA, but this is not a verified per-rail budget. The 0.5 A ASIC feed allocation is a provisional allowance inside the 3.3 V rating, not an additional 0.5 A to add again.

One group of seven LT3042s contributes approximately 14 mA typical light-load overhead, or 46.2 mW at 3.3 V. Eight groups would contribute approximately 369.6 mW before ASIC load. Sharing may reduce overhead, but acceptable noise and transient behavior must be demonstrated.

To determine actual FPGA current, use AMD power analysis after implementing capture/GPIF/optional DDR with realistic switching activity and I/O loads, then measure. Data rate alone does not reveal internal switched capacitance, clock activity or temperature.

Sources: DS181 [S19], Nexys power provision [S03], LT3042 [S18], Gerald slide 7 [S01].

Where the VDD10 / VSS10 concern comes from

Location: **Original Sources / 04_LDO_Routing_Reference / LDO_Board_10SOIC (New Rigid) / PCB.kicad_sch**. This is the supplied LDO reference, not the Nexys design or a newly designed circuit.

Circuit in that file	What was found	Meaning
U8 + R9	LT3042; SET resistor R9 = 90 kohm; local ground = AGND; output named VDD10	Nominal output = $100\text{ }\mu\text{A} \times 90\text{ kohm} = 9\text{ V}$ above AGND. The net name does not set the voltage.
U9 + R8	R8 = 90 kohm; regulator GND/EP tied to INPUT_VSS10; IN tied to AGND	If regulating as drawn, $V_{OUT} = INPUT_VSS10 + 9\text{ V}$. A -10 V input would imply -1 V output, not -10 V.
J15 / J8	External INPUT_VDD10 / INPUT_VSS10 supply connections	The shown regulators do not create the positive and negative input supplies.

LT3042 sets its output relative to its **own GND terminal**. That is why a floating ground changes the calculation. This observation does not prove the lab hardware is wrong: it may use another revision, resistor value, wiring or package. It proves we need that actual revision before copying.

Question for Gerald: What positive/negative voltages are required at the ASIC pins, which supplies feed U8/U9, and is this exact drawing the tested revision? Please provide measured voltages and the assembled resistor values.

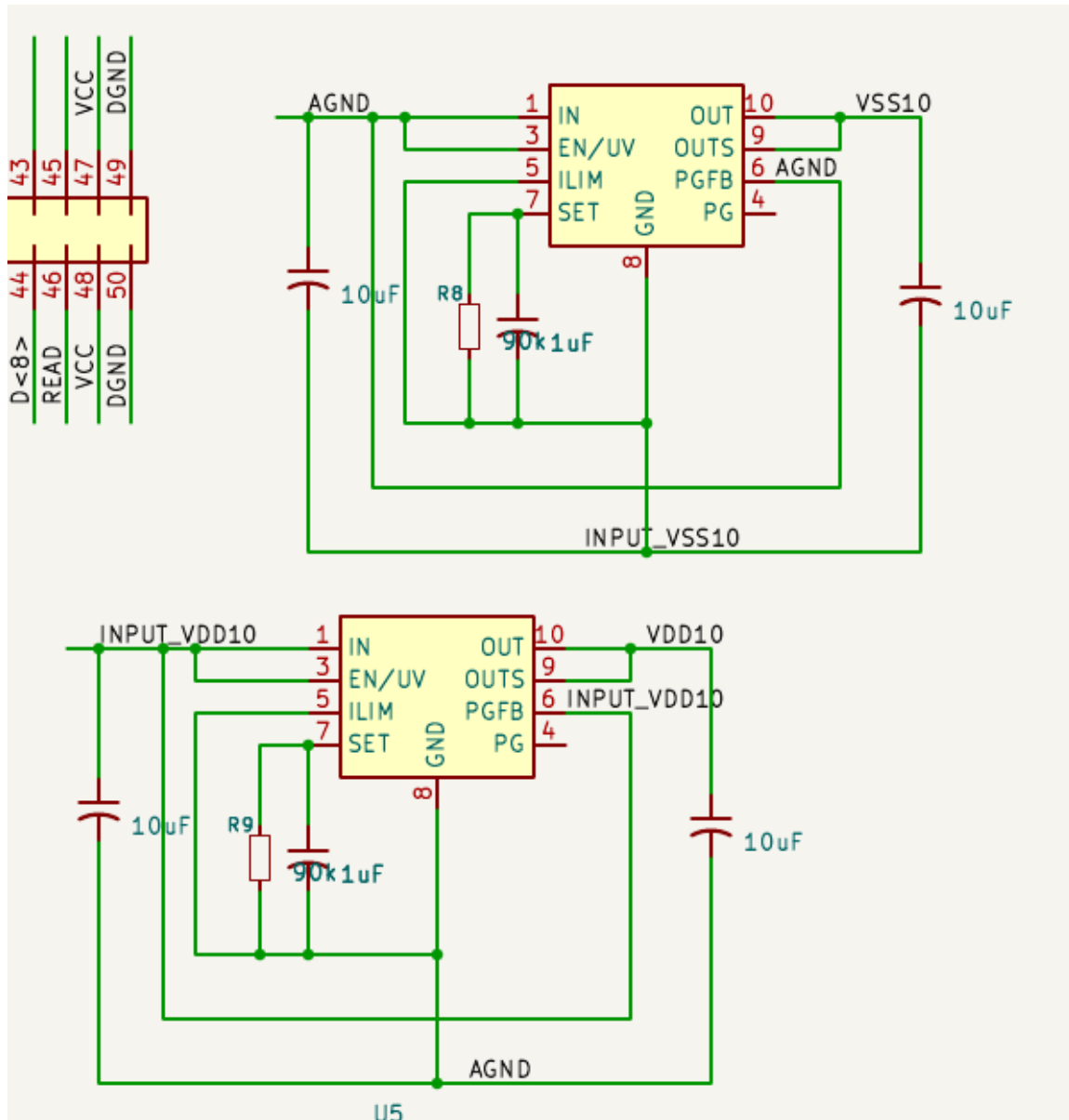
Additional footprint issue: the old schematic uses LT3042xMSE symbol pin numbering but assigns a DFN-10 footprint. The manufacturer MSOP and DFN pin maps differ. Ask for the actual assembled order code and corrected symbol/footprint mapping; do not assume a symbol name can be changed without checking every pin.

The original source excerpt is reproduced on the following page. It is source evidence, not a replacement schematic.

Sources: supplied schematic U8/R9 and U9/R8 [S17]; LT3042 SET equation and package pinouts [S18].

Original schematic excerpt: the two rails

Source: the single A3 root sheet, page 1, upper-right area. **Top: U9/R8, VSS10. Bottom: U8/R9, VDD10.** U8/U9 reference labels are hidden in the original; R8/R9 identify the branches.



This excerpt is preserved from the supplied schematic. Some original labels overlap. Both SET resistors are 90 kohm in the actual source objects; values were checked from the source/netlist as well as the drawing.

Bring this page to Gerald: confirm the tested revision, actual resistor values, expected input/output voltages and the LT3042 package/pin mapping.

Sources: original KiCad file [S17]; manufacturer SET equation and pin maps [S18]. This is not proof that an assembled board is faulty.

Clocks, programming and pin planning

Function	Physical parts / count	What it means
FPGA reference clock	1 active oscillator; propose 100 MHz + 1 local bypass capacitor	A small powered part produces a stable square wave. FPGA clock blocks derive internal/ASIC clocks from it.
FX3 reference clock	1 supported crystal; propose 19.2 MHz + 2 load capacitors	The USB controller needs its own reference. A crystal is passive and works with the chip oscillator circuit. Verify exact load network.
GPIF clock	1 clock signal between FPGA and FX3	Not necessarily a third oscillator. For synchronous slave FIFO, define its source, direction and phase/timing explicitly.
JTAG access	1 programming header with matching external programmer	Connector carries TCK/TMS/TDI/TDO, reference voltage and ground. Standard headers may have extra ground/control contacts.
Boot/reset support	Mode straps, required pull-ups, reset control and status connections; exact count open	Resistors define boot behavior; PROGRAM_B can reload FPGA, INIT_B/DONE expose status. These are more than a connector.

FX3 is not automatically a Xilinx-compatible JTAG programmer. Custom firmware can implement a programming path, but it needs software, wiring and validation. Keep an independent JTAG recovery connection in the first custom board; an onboard USB-JTAG bridge is not necessary if we use an external programmer.

The SBG484 device has **285 user I/O**: bank 13 has 35, and banks 14/15/16/34/35 have 50 each. Configuration bank 0 is separate. Bank 14 includes QSPI signals. Do not assign 1.5 V ASIC I/O to a bank that must support a 3.3 V flash/interface.

Build a Vivado pin-and-clock skeleton before final connectors. A large pin total is not proof that returned clocks, GPIF and DDR can use any pins. Nexys FMC exposes only a subset; it may not fit all eight ASICs and FX3 simultaneously.

Nexys supplies the test FPGA and convenient known-working circuits. Our PCB needs its own chosen clocks, power, flash, connectors and layout.

Sources: AMD configuration and package guidance [S04-S06], FX3 [S08-S09], Nexys [S03].

Firmware and receiver: work you can own

Layer	Work to implement	Evidence it works
FPGA: ASIC control	Clock/reset sequencing, gain/stimulation SPI, safe defaults and status	Measured pins and documented ASIC response
FPGA: acquisition	Capture 64 serial outputs, returned-clock timing, READ/SYNC and channel identity	Known channel patterns, one ASIC before scaling
FPGA: transport	Clock-domain FIFOs, packet format, sequence/CRC, DDR if selected, GPIF flow control	Simulated stalls/errors, timing closure, hardware counters
FX3 firmware	GPIF state machine, DMA, USB endpoints, queued transfers and command path	Sustained generated data without lost/duplicated words
KR260 receiver	Asynchronous USB reads, ring buffers, packet/error checks and forwarder	Measured throughput, queue depth, stalls and CPU use
KR260 network system	10 GbE/SFP+ hardware path, drivers/configuration, compatible cable/modules	Full USB reception and network forwarding concurrently
PC recorder/viewer	Reliable file writer, metadata/error log, command UI and selected-channel display	Saved data matches source; storage keeps up

Commands return from PC/KR260 through USB to FX3 and FPGA, then to ASIC control pins. A successful USB write only means a transfer was accepted at that layer; it does not prove the ASIC executed a command. Add acknowledgements/readback/status where supported.

The PC needs approximately 950.4 GB/hour at the current framing. A 1 GbE link is insufficient. A 10 GbE physical interface has enough headline rate, but hardware, driver, CPU, transport and storage still determine useful throughput.

We can implement and test the receiver ourselves. Gerald supplies ASIC-specific facts; vendor datasheets establish component limits; our bench testing establishes actual end-to-end performance.

Sources: reference implementation [S02], FX3 examples [S09], KR260 [S13].

What must be finished before ordering?

Gate	Concrete work / owner	Result required
1. Source identity	Gerald supplies current ASIC/carrier/LDO revisions and assembled package/resistor information	Versioned pin map and corrected source circuits
2. Electrical contract	Gerald supplies ASIC voltage/timing/current limits; we check datasheets and measure missing currents	Every connector contact has direction, voltage, timing and reference ground
3. Link proof	We implement generated stream -> FX3 -> KR260 -> PC	Agreed rate plus margin, no silent loss, measured maximum stalls
4. Memory and pins	We choose buffer duration, controller and legal bank/clock assignments	Timing-valid top-level project and selected RAM architecture
5. Power circuit	We complete converter calculations, all support counts, sequencing, noise and thermal checks	Exact rail schematic and component ratings/footprints
6. BOM and layout	We reconcile symbols, footprints, mating connectors, stackup and PCB checks	Orderable BOM from schematic; reviewed ERC/DRC and assembly files
7. Bring-up plan	We document safe startup, measurements, test patterns and recovery	Rails -> JTAG -> clocks/flash -> transport -> one ASIC -> eight ASICs

Startup matters: the recommended core/BRAM capacitors total 844.39 uF. Charging through 1 V in 5 ms draws about 169 mA just for those capacitors. DS181 power-on-current expressions evaluated with its typical quiescent values add about 759 mA. This illustrative 0.928 A is not a maximum; actual ramp, extra capacitors and silicon conditions matter.

ADP5052 12 V -> 1 V conversion also constrains switching frequency through minimum on-time. Check input transients and simultaneous package heating. FX3 PHY startup needs transient-capable decoupling. Follow AMD supply sequencing and unused-pin guidance.

The next pages are the questions to bring to Gerald. They ask for specific facts and artifacts, rather than a general "does this look okay?"

Sources: power/current/sequencing [S19], ADP5052 [S14], FX3 hardware [S08].

Questions for Gerald | 01-04

Bring current source files or measured values where possible. "Unknown - measure on board X" is an acceptable answer with a named next step.

G01. Which exact ASIC and carrier revisions are current, and which files are authoritative?

Bring / answer: Revision/date, released schematic, pin map, known changes from slides/repository.

Why: Avoid designing to an old revision.

Owner / source: Gerald / ASIC designer; versioned design archive.

Answer / file / next step: _____

G02. Can we get the connector-to-ASIC contact table and exact mating connector/orientation?

Bring / answer: Contact number, signal, ASIC identity, direction, power/ground role, part numbers and pin-1 drawing.

Why: Closes connector and routing correctness.

Owner / source: Gerald + carrier designer; CAD/pin sheet, then continuity check.

Answer / file / next step: _____

G03. Is 1.5 V digital I/O correct for this ASIC revision, and what are its guaranteed limits?

Bring / answer: Nominal/tolerance, VIH/VIL, VOH/VOL, absolute maximum, behavior while unpowered.

Why: Selects FPGA bank voltage and any level translation/isolation.

Owner / source: ASIC designer specification; Gerald identifies the source.

Answer / file / next step: _____

G04. What clock-to-data, setup/hold, edge and duty-cycle limits apply at the carrier connector?

Bring / answer: Timing diagram with minimum/maximum values and returned-clock relationship.

Why: Enables FPGA timing constraints and PCB skew budget.

Owner / source: ASIC specification/characterization; our team measures the implementation.

Answer / file / next step: _____

Questions for Gerald | 05-08

G05. Has this ASIC/carrier been validated at 32 MHz / 31.25 kS/s/channel, and why is reference code at 16 MHz?

Bring / answer: Tested modes, measured sample rates and reason for lower-rate code.

Why: Separates target bandwidth from already-demonstrated silicon operation.

Owner / source: Gerald / hardware-test owner; test logs and reference firmware.

Answer / file / next step: _____

G06. What reset, clock-start and gain-programming sequence is required?

Bring / answer: Polarities, pulse lengths, startup order, settling delay and known-working configuration.

Why: Determines boot logic and safe pull levels.

Owner / source: Gerald / ASIC designer; operating procedure plus bench verification.

Answer / file / next step: _____

G07. Which clocks/resets/programming/stimulation signals can be shared across eight ASICs?

Bring / answer: Mark each shared/per-chip; fanout limit and inter-chip synchronization tolerance.

Why: Closes total signal count and clock buffering requirements.

Owner / source: Gerald for experiment; ASIC designer for electrical limits; we design buffers.

Answer / file / next step: _____

G08. Where are the left/right programming map, gain calibration and AC_IN/IMP_TEST pin specifications?

Bring / answer: Bit/register tables, legal defaults, gain table, auxiliary pin role/range/unused termination.

Why: Prevents sending correctly timed but semantically wrong settings.

Owner / source: Gerald / ASIC designer; configuration files and pin specifications.

Answer / file / next step: _____

Questions for Gerald | 09-12

G09. Are five 1.5 V rails, VDD_CGEN=1.6 V and REF_VCM=0.75 V correct for the current chip?

Bring / answer: Each rail name, approved voltage/tolerance and load served.

Why: Confirms the resistor-derived setpoints before reuse.

Owner / source: Gerald / ASIC designer; validated schematic and specification.

Answer / file / next step: _____

G10. Do we have typical and peak current per rail in reset, recording and stimulation?

Bring / answer: Per-ASIC table with clock, channels and operating condition; if absent, identify board/test points.

Why: Allows regulators, traces and thermal estimates to be sized.

Owner / source: Existing characterization/simulation, or measurements our team performs.

Answer / file / next step: _____

G11. What noise/ripple, startup ramp/order and shutdown requirements apply?

Bring / answer: Numerical noise limit and bandwidth; allowed rail differences, sequencing and ramp limits.

Why: Defines regulator/filter design and a measurable acceptance test.

Owner / source: ASIC designer requirements; our team verifies the power system.

Answer / file / next step: _____

G12. Should VDD10 be +9 V or +10 V, and is R9 really 90 kohm on the tested board?

Bring / answer: Approved VDD10 and INPUT_VDD10 values, assembled R9 value and tested revision.

Why: The provided U8 SET calculation gives +9 V despite its name.

Owner / source: Gerald / regulator-board designer; actual measurement and released files.

Answer / file / next step: _____

Questions for Gerald | 13-16

G13. What should INPUT_VSS10 and VSS10 be, and is U9/R8 the validated topology?

Bring / answer: Input/output voltages, current direction, known-working schematic and measured output.

Why: The floating positive LDO does not simply create -10 V.

Owner / source: Gerald / regulator designer; LT3042 analysis and bench check.

Answer / file / next step: _____

G14. Which exact LT3042 package was assembled, and is there a corrected symbol-to-pad map?

Bring / answer: Complete part number, MSOP/DFN package, pin 1-10/EP mapping, released PCB/BOM.

Why: Old symbols use MSE numbering while footprint fields specify DFN.

Owner / source: Gerald / board designer; assembly records and ADI package pinouts.

Answer / file / next step: _____

G15. Which input feed is intended, and where are AGND/DGND/shields joined?

Bring / answer: Resolve old +2 V versus 3V3 input; give voltage tolerance and exact ground junction/location.

Why: Avoids combining incompatible references or incomplete return paths.

Owner / source: Gerald identifies working pairing; we check full schematic and continuity.

Answer / file / next step: _____

G16. Which rails must be local/per-chip, and which may share regulators?

Bring / answer: Allowed groups, supporting noise/crosstalk results and an agreed test if unknown.

Why: Determines whether 7, 28 or 56 low-voltage LDOs are needed and their overhead.

Owner / source: Gerald for signal-quality goals; ASIC designer and our measurements for sharing.

Answer / file / next step: _____

Questions for Gerald | 17-20

G17. What concurrent stimulation must the first board support?

Bring / answer: Active outputs per chip/stack, maximum current, pulse duration, repetition and duty cycle.

Why: Defines peak and average supply load; chip capability alone is insufficient.

Owner / source: Gerald / experiment owner; we calculate and test resulting power.

Answer / file / next step: _____

G18. Which configuration files and STIM_EN/CLK/START/CHB waveforms are known to work?

Bring / answer: Example bitstreams/spreadsheets, pulse count and expected timing traces.

Why: Verifies actual control semantics rather than relying on comments.

Owner / source: Gerald / original control author; our FPGA implementation reproduces it.

Answer / file / next step: _____

G19. What samples are invalid during/after stimulation, and how long until recovery?

Bring / answer: Reset/blanking timing, valid-data delay and required metadata marking.

Why: A continuous USB stream does not mean continuously valid neural data.

Owner / source: Gerald / ASIC characterization; our firmware marks and verifies data.

Answer / file / next step: _____

G20. What states are required at power-up, lost configuration, stopped clock, cable disconnect and reset?

Bring / answer: Control levels, electrode/stimulation state and charge-balance requirements.

Why: Determines pull resistors and interlocks; validate on dummy loads.

Owner / source: Gerald / ASIC designer operating procedure; our team tests behavior.

Answer / file / next step: _____

Questions for Gerald | 21-24

G21. Which carrier dimensions and keep-outs must remain despite flexible board size?

Bring / answer: Connector position/height, wire-bond clearance, thickness, mounts and flex bend limits.

Why: Keeps the new board compatible with the existing stack.

Owner / source: Carrier designer/Zitong + Gerald; CAD and measured hardware.

Answer / file / next step: _____

G22. How long must a recording run, and what should happen if a downstream stall fills the buffer?

Bring / answer: Required run duration; permitted losses if any; stop/flag/drop behavior and maximum tolerated gap.

Why: Determines RAM capacity, error handling and storage requirements.

Owner / source: Gerald/experiment owner defines behavior; our team measures actual stalls.

Answer / file / next step: _____

G23. How accurately must all ASICs and any external experiment trigger be aligned?

Bring / answer: Allowed inter-chip/time-stamp error, trigger voltage/direction and event timing requirements.

Why: Determines clock/trigger connectors, synchronization logic and timestamp format.

Owner / source: Gerald/experiment owner; ASIC timing sources and our implementation tests.

Answer / file / next step: _____

G24. What measurable results define a successful eight-ASIC bench demonstration?

Bring / answer: Full-rate duration, allowed frame errors, gain/channel/noise checks and required stimulation tests.

Why: Turns completion into a repeatable test rather than a subjective impression.

Owner / source: Gerald agrees experiment criteria; our team supplies test results.

Answer / file / next step: _____

Sources and evidence | 1

Manufacturer links are clickable. Local originals are preserved in the FPGA PCB folder; their filenames identify the evidence to bring to colleagues. Quantities in this packet are preliminary, not purchase approval.

[S01] [Gerald: Chip_FPGA Interface.pptx](#)

Recording target, power, ASIC control and stimulation capabilities; not guaranteed electrical limits.

Original Sources/02_ASIC_Interface_Slides/Chip_FPGA Interface.pptx

[S02] [FPGA_512 reference repository](#)

Reviewed snapshot 767e82528780005cbcb37b3e926197755bac622e; ECP5/FT600 reference, not current Artix/FX3 hardware proof.

[S03] [Digilent Nexys Video reference manual](#)

Same FPGA, test-board rails/FMC/DDR. Reference/test equipment, not final PCB.

[S04] [AMD UG470 configuration guide](#)

Bitstream size, SPI boot, configuration pins and startup.

[S05] [AMD DS180 device overview](#)

XC7A200T block RAM and device resources.

[S06] [AMD UG475 packages and pinouts](#)

Use with archived xc7a200tsbg484pkg.csv and UG471 to close pin/bank assignments.

[S07] [AMD supported Artix-7 configuration memories](#)

Check exact flash alias/order code and addressing settings.

[S08] [Infineon CYUSB301X / CYUSB201X datasheet](#)

Clock, voltage domains, USB/GPIF capabilities and hardware support.

[S09] [Infineon AN65974 slave FIFO interface](#)

32-bit synchronous GPIF, DMA/flow control and external FPGA examples.

[S10] [Microchip SuperSpeed USB explanation](#)

TX/RX pairs, contact roles and 8b/10b coding.

Sources and evidence | 2

Manufacturer links are clickable. Local originals are preserved in the FPGA PCB folder; their filenames identify the evidence to bring to colleagues. Quantities in this packet are preliminary, not purchase approval.

[S11] [Wurth 692221030100 manufacturer drawing](#)

9-contact USB3 Type-B candidate; mechanical/footprint source.

[S12] [TI HD3SS3220 Type-C controller/mux](#)

Optional CC/orientation circuit; Type-C does not add FX3 lanes.

[S13] [AMD KR260 UG1092 interfaces](#)

USB host topology and network interfaces.

[S14] [ADI ADP5052 datasheet](#)

Figure 55 and design sections; exact topology counts, timing/compensation must be recalculated.

[S15] [TI TPS62160 datasheet](#)

Figure 39, 1.2 V / 1 A reference and component guidance.

[S16] [AMD UG483 PCB and decoupling](#)

Table 2-2 XC7A200T/SBG484 and Table 2-5 component electrical guidance.

[S17] Supplied LDO schematic

Single root sheet. U8/R9 and U9/R8 voltage reconstruction; package mismatch is a source-file finding.

Original Sources/04_LDO_Routing_Reference/LDO_Board_10SOIC (New Rigid)/PCB.kicad_sch

[S18] [ADI LT3042 datasheet](#)

100 uA SET current, 200 mA capacity, support network and different MSOP/DFN pin maps.

[S19] [AMD DS181 Artix-7 electrical limits](#)

Rails/startup/ramp limits. Typical quiescent values use nominal voltage, 85 C junction, blank configured device.

[S20] [Vishay Si7232DN datasheet](#)

One package contains two N-channel MOSFETs; voltage/transient review required.

Sources and evidence | 3

Manufacturer links are clickable. Local originals are preserved in the FPGA PCB folder; their filenames identify the evidence to bring to colleagues. Quantities in this packet are preliminary, not purchase approval.

[S21] [SiTime SiT8008B oscillator datasheet](#)

100 MHz family candidate, bypass and OE guidance; full order code open.

[S22] [Infineon Explorer Kit CYUSB3KIT-003](#)

One bench kit contains FX3, connector and support circuitry. FMC adapter compatibility remains unverified.

[S23] [Infineon GPIF-to-USB performance clarification](#)

Historical vendor ~375 MB/s capability statement; not a measured KR260 guarantee.

[S24] [Infineon AN86947 throughput](#)

Internal GPIF/AUTO-DMA test is not an external FPGA-to-KR260 benchmark; USB-only figures are not GPIF throughput.

[S25] [Infineon FPGA programming through FX3](#)

Custom firmware/slave-serial example; FX3 is not an automatic Vivado JTAG programmer.

[S26] [Original meeting and lab conversation](#)

Boot-memory discussion and system context. Slack receipt date is not verified message chronology.

9_17_26/FPGA_PCB_Meeting_Notes_2026-09-17.docx; Original Sources/10_Lab_Slack/

[S27] [Infineon S25FL256SAGMFI000 candidate](#)

Full flash candidate part; package and configuration must be checked before ordering.

[S28] [TE 1-2295018-2 Type-C alternative](#)

24-contact alternative; would replace Type-B, not add a required second acquisition port.